Infronprompt-cache-bench · reproducible inference benchmark artifact

Infron vs OpenRouter Routing, Cache, Cost, and Streaming Performance A/B Test Report

Abstract and Executive Outline

Keywords: Prompt Caching; A/B Testing; Provider Routing; Cache Affinity; Latency; Throughput; Cost Attribution; gemini-3-flash-preview

Abstract

This report evaluates `google/gemini-3-flash-preview` on Infron and OpenRouter across cache reuse, observed cost, throughput, E2E latency, and Streaming TTFT under Prompt Caching workloads.

The main findings are: Infron and OpenRouter tie on Cache hit rate in all routing modes; Infron leads Observed cost in 1/4 routing modes; OpenRouter leads in 3/4; 0/4 tie; Infron leads Throughput in all routing modes; Infron leads E2E latency in all routing modes; Infron leads Streaming TTFT in all routing modes.

Overall, Infron's cross-mode strengths are throughput, Streaming TTFT, E2E latency, while OpenRouter's cross-mode strengths are observed cost. Platform choice should be driven by workload objective rather than a single headline metric.

Figure 0: Normalized Capability Radar

The five radar axes represent throughput, price, E2E latency, Streaming TTFT, and cache hit rate. Every metric is normalized to a 0–100 score, with farther outward meaning better.

Thick solid lines show platform-level contours; translucent lines and points show individual routing modes.

Conclusion Overview: Core Metric Winners by Routing Mode

Based on strict A/B paired samples. Blue represents Infron, orange represents OpenRouter; gold cells mark the winner for the routing objective.

Throughput	Observed cost	E2E latency	Streaming TTFT	Cache hit rate
Infron wins 4/4 Max advantage 4.83%, higher is better	OpenRouter wins 3/4 Max advantage 28.85%, lower is better	Infron wins 4/4 Max advantage 42.39%, lower is better	Infron wins 4/4 Max advantage 36.72%, lower is better	Tie in 4/4 modes Max advantage 0.00%, higher is better

Routing mode	Throughput objective	Cost objective	Latency objective	TTFT objective	Cache result
Throughput First throughput	Infron advantage 4.70%	OpenRouter advantage 3.31%	Infron advantage 42.39%	Infron advantage 34.14%	Tie tie
Price First price	Infron advantage 4.50%	OpenRouter advantage 3.35%	Infron advantage 37.65%	Infron advantage 35.04%	Tie tie
Latency First latency	Infron advantage 4.30%	OpenRouter advantage 3.35%	Infron advantage 32.48%	Infron advantage 32.07%	Tie tie
TTFT First ttft	Infron advantage 4.83%	Infron advantage 28.85%	Infron advantage 37.84%	Infron advantage 36.72%	Tie tie

Executive Outline

Dimension	Conclusion	Evidence
Controls	First/second `usage.prompt_tokens` deltas are limited to 50 tokens within each `sort/group/round` pair.	Methods and data quality
Cache reuse	Infron and OpenRouter tie on Cache hit rate in all routing modes	Overall metrics and mechanism section
Observed cost	Infron leads Observed cost in 1/4 routing modes; OpenRouter leads in 3/4; 0/4 tie	Overall metrics and provider drill-down
Performance	Infron leads Throughput in all routing modes; Infron leads E2E latency in all routing modes; Infron leads Streaming TTFT in all routing modes	Charts and statistical tests
Attribution boundary	Claims use observable response telemetry: usage, cost, TTFT, latency, provider fields, and cache tokens.	Provider/Route drill-down
Business meaning	Long-context, RAG-prefix, agent-tool, and high-frequency template workloads should evaluate cache rate, cost, first-token latency, and E2E latency together.	Discussion and conclusion

Routing-Mode Conclusions

Routing mode	Objective	Cache winner	Cost winner	Throughput winner	E2E latency winner	TTFT winner	Interpretation
Throughput First	Maximize output capacity per unit time	**Tie**	**OpenRouter**	**Infron**	**Infron**	**Infron**	Infron leads overall (3/5 metrics)
Price First	Minimize request and token cost	**Tie**	**OpenRouter**	**Infron**	**Infron**	**Infron**	Infron leads overall (3/5 metrics)
Latency First	Minimize full-response waiting time	**Tie**	**OpenRouter**	**Infron**	**Infron**	**Infron**	Infron leads overall (3/5 metrics)
TTFT First	Minimize streaming first-token time	**Tie**	**Infron**	**Infron**	**Infron**	**Infron**	Infron leads overall (4/5 metrics)

1. Introduction: Background, Questions, and Contributions

LLM inference-platform behavior is shaped by provider routing, prompt caching, streaming response handling, cost attribution, and fallback policy. This report treats the platform as an observable system and measures speed, cost, cache reuse, and first-token experience through strict A/B pairs.

1.1 Research Hypotheses

Hypothesis	Statement	Validation metric
H1	Stronger provider/cache affinity improves token-level cache hit rate for repeated stable prefixes.	Second-request cache-read tokens and token cache hit rate
H2	Higher cache hit rate can reduce observed cost, but does not necessarily reduce TTFT or E2E latency.	Observed cost, average TTFT, average request latency
H3	Different routing sorts change provider selection and produce different cost/throughput/latency frontiers.	Provider distribution, throughput, latency, cost

1.2 Contributions

- Uses response-returned `usage.prompt_tokens` as the input-token control while allowing small cross-platform accounting variance up to 50 tokens.
- Extends prompt-caching evaluation to cost, throughput, E2E latency, TTFT, provider distribution, reasoning telemetry, and paired statistical tests.
- Scopes every claim to observable response telemetry instead of private routing internals.

2. Experimental Design, Dataset, and Controls

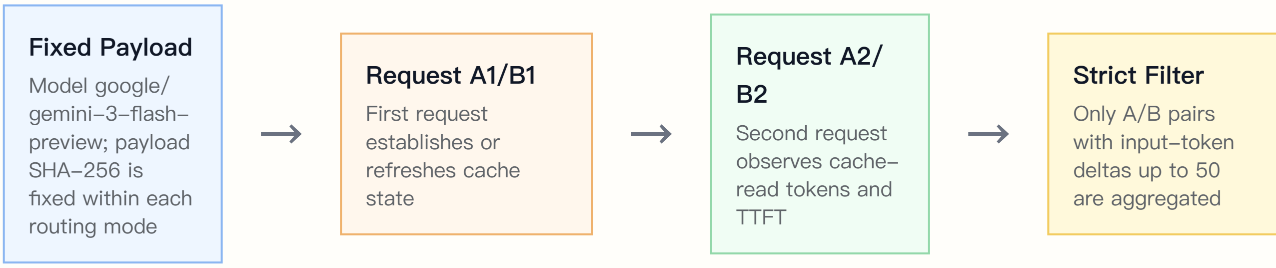
2.1 Dataset Construction

The dataset is `controlled_cache_probe`, covering 4 routing sorts, 2 platforms, 4 groups, and 50 rounds per group. Each round sends identical first/second Chat Completions requests; the second request observes cache-read tokens, TTFT, and E2E latency.

The built-in business templates represent stable long-context workloads such as RAG support, agent tool instructions, marketing automation, and code review.

2.2 Controlled Variables

Figure 1: Experimental Design and Strict A/B Pairing Filter



Controlled-variable rule: within the same `sort/group/round`, both platforms must have first/second `usage.prompt_tokens` deltas no greater than 50 tokens. Total Input Tokens are computed from response-returned usage, not local tokenizer estimates.

2.3 Metric Definitions

Metric	Definition	Direction
Call cache hit rate	Share of second requests with `cache_read_tokens > 0`	Higher is better
Token cache hit rate	Second-request cache-read tokens / second-request prompt tokens	Higher is better
Observed cost	Sum of first + second request usage/cost values	Lower is better
Throughput	Completion tokens / request latency seconds	Higher is better
E2E latency	Full response latency per request	Lower is better
TTFT	Streaming first-token arrival time	Lower is better
Reasoning telemetry	Reasoning tokens from response usage	Used to explain latency, throughput, and cost

3. Experimental Environment and Data Quality

Item	Configuration
Model	`google/gemini-3-flash-preview`
Provider model IDs	`google/gemini-3-flash-preview`
Platforms	Infron and OpenRouter
API protocol	`/v1/chat/completions`
Routing modes	Throughput First, Price First, Latency First, TTFT First
Groups	4
Rounds per group	50
Workers	24
Request mode	Streaming Chat Completions with TTFT collection
Reasoning / thinking control	No explicit reasoning/thinking parameter; model and platform defaults are preserved
Prompt length tiers	Not enabled
Excluded records	16

4. Results: Overall Metrics and Main Findings

Routing mode	Platform	Strict pairs	Total Input Tokens	Token cache hit rate	Observed cost	Throughput	E2E latency	Streaming TTFT
Throughput First	**Infron**	192	619007	**0.00%**	$0.32311900	**7.79 tok/s**	**1540.55 ms**	**1472.19 ms**
Throughput First	OpenRouter	192	618628	**0.00%**	**$0.31277000**	1.37 tok/s	2193.61 ms	1974.82 ms
Price First	**Infron**	200	644800	**0.00%**	$0.33680000	**7.28 tok/s**	**1649.17 ms**	**1580.24 ms**
Price First	OpenRouter	200	644556	**0.00%**	**$0.32587800**	1.32 tok/s	2270.02 ms	2134.00 ms
Latency First	**Infron**	200	644800	**0.00%**	$0.33680000	**7.28 tok/s**	**1648.06 ms**	**1579.91 ms**
Latency First	OpenRouter	200	644592	**0.00%**	**$0.32589600**	1.37 tok/s	2183.36 ms	2086.62 ms
TTFT First	**Infron**	200	644400	**0.00%**	**$0.25294200**	**7.48 tok/s**	**1694.77 ms**	**1588.25 ms**
TTFT First	OpenRouter	200	644650	**0.00%**	$0.32592500	1.28 tok/s	2336.14 ms	2171.44 ms

4.1 Tail Latency and Statistical Tests

Tail percentiles expose risk that averages hide.

Routing mode	Platform	P50 latency	P95 latency	P99 latency	P50 TTFT	P95 TTFT	P99 TTFT
Throughput First	**Infron**	**1512.71 ms**	**1739.76 ms**	**2260.72 ms**	**1442.41 ms**	**1676.22 ms**	**2190.41 ms**
Throughput First	OpenRouter	2062.09 ms	3012.83 ms	3775.55 ms	1894.22 ms	2601.23 ms	3096.72 ms
Price First	**Infron**	**1629.73 ms**	**1780.04 ms**	**2329.37 ms**	**1562.17 ms**	**1704.90 ms**	**2271.10 ms**
Price First	OpenRouter	2199.30 ms	2769.94 ms	3415.59 ms	2015.11 ms	2606.43 ms	3392.62 ms
Latency First	**Infron**	**1632.59 ms**	**1805.49 ms**	**2292.44 ms**	**1563.95 ms**	**1741.77 ms**	**2190.49 ms**
Latency First	OpenRouter	2145.18 ms	2759.26 ms	3536.06 ms	2022.17 ms	2640.24 ms	3486.98 ms
TTFT First	**Infron**	**1656.76 ms**	**2027.84 ms**	**2495.79 ms**	**1560.16 ms**	**1963.87 ms**	**2409.80 ms**
TTFT First	OpenRouter	2207.73 ms	3100.51 ms	6477.73 ms	2073.13 ms	2974.58 ms	4397.60 ms

Mean deltas use bootstrap 95% CIs; p-values use paired sign-flip permutation tests.

Routing mode	Metric	Mean delta	95% CI	p-value	Pairs	Interpretation
Throughput First	Latency: OpenRouter – Infron	**1306.12 ms**	1209.85 ms to 1411.80 ms	<0.001	192	Positive means lower Infron latency
Throughput First	TTFT: OpenRouter – Infron	**1005.25 ms**	930.03 ms to 1085.87 ms	<0.001	192	Positive means lower Infron TTFT
Throughput First	Throughput: Infron – OpenRouter	**6.4449 tok/s**	6.3601 tok/s to 6.5274 tok/s	<0.001	192	Positive means higher Infron throughput
Throughput First	Cost: OpenRouter – Infron	**$–0.00005390**	$–0.00005500 to $–0.00005172	<0.001	192	Positive means lower Infron cost
Throughput First	Token Cache Hit: Infron – OpenRouter	0.00 pp	0.00 pp to 0.00 pp	1.0000	192	Positive means higher Infron cache hit
Price First	Latency: OpenRouter – Infron	**1241.70 ms**	1102.82 ms to 1446.43 ms	<0.001	200	Positive means lower Infron latency
Price First	TTFT: OpenRouter – Infron	**1107.53 ms**	966.72 ms to 1316.38 ms	<0.001	200	Positive means lower Infron TTFT
Price First	Throughput: Infron – OpenRouter	**5.9548 tok/s**	5.8851 tok/s to 6.0253 tok/s	<0.001	200	Positive means higher Infron throughput
Price First	Cost: OpenRouter – Infron	**$–0.00005461**	$–0.00005470 to $–0.00005451	<0.001	200	Positive means lower Infron cost
Price First	Token Cache Hit: Infron – OpenRouter	0.00 pp	0.00 pp to 0.00 pp	1.0000	200	Positive means higher Infron cache hit
Latency First	Latency: OpenRouter – Infron	**1070.61 ms**	991.57 ms to 1157.04 ms	<0.001	200	Positive means lower Infron latency
Latency First	TTFT: OpenRouter – Infron	**1013.43 ms**	931.35 ms to 1101.19 ms	<0.001	200	Positive means lower Infron TTFT

Routing mode	Metric	Mean delta	95% CI	p-value	Pairs	Interpretation
Latency First	Throughput: Infron – OpenRouter	**5.9127 tok/s**	5.8543 tok/s to 5.9752 tok/s	<0.001	200	Positive means higher Infron throughput
Latency First	Cost: OpenRouter – Infron	**$–0.00005452**	$–0.00005462 to $–0.00005441	<0.001	200	Positive means lower Infron cost
Latency First	Token Cache Hit: Infron – OpenRouter	0.00 pp	0.00 pp to 0.00 pp	1.0000	200	Positive means higher Infron cache hit
TTFT First	Latency: OpenRouter – Infron	**1282.75 ms**	1124.74 ms to 1450.04 ms	<0.001	200	Positive means lower Infron latency
TTFT First	TTFT: OpenRouter – Infron	**1166.39 ms**	1039.44 ms to 1307.90 ms	<0.001	200	Positive means lower Infron TTFT
TTFT First	Throughput: Infron – OpenRouter	**6.1956 tok/s**	6.1064 tok/s to 6.2779 tok/s	<0.001	200	Positive means higher Infron throughput
TTFT First	Cost: OpenRouter – Infron	**$0.00036492**	$0.00036471 to $0.00036513	<0.001	200	Positive means lower Infron cost
TTFT First	Token Cache Hit: Infron – OpenRouter	0.00 pp	0.00 pp to 0.00 pp	1.0000	200	Positive means higher Infron cache hit

4.2 Reasoning / Thinking Control Check

This run does not explicitly set reasoning/thinking parameters and keeps model/platform defaults; this table records reasoning telemetry under default behavior.

Routing mode	Platform	Reasoning tokens	Avg reasoning tokens/request	Reasoning requests	Avg first reasoning token	Avg TTFT	Avg E2E latency
Throughput First	**Infron**	4293	11.1797	384	**0.00 ms**	**1472.19 ms**	**1540.55 ms**
Throughput First	OpenRouter	**0**	**0.0000**	**0**	2173.87 ms	1974.82 ms	2193.61 ms
Price First	**Infron**	4441	11.1025	400	**0.00 ms**	**1580.24 ms**	**1649.17 ms**
Price First	OpenRouter	**0**	**0.0000**	**0**	2253.46 ms	2134.00 ms	2270.02 ms
Latency First	**Infron**	4447	11.1175	400	**0.00 ms**	**1579.91 ms**	**1648.06 ms**
Latency First	OpenRouter	**0**	**0.0000**	**0**	2168.73 ms	2086.62 ms	2183.36 ms
TTFT First	**Infron**	5071	12.6775	400	**0.00 ms**	**1588.25 ms**	**1694.77 ms**
TTFT First	OpenRouter	**0**	**0.0000**	**0**	2319.20 ms	2171.44 ms	2336.14 ms

4.3 API Protocol Compatibility Matrix

This run uses `/v1/chat/completions` as the API protocol.

5. Result Visualizations by Routing Mode

The following charts are driven by the strict-paired summary data and present routing-mode, platform, cost, cache, E2E latency, Streaming TTFT, and upstream provider differences.

5.1 Core Metric Chart Overview

The charts below are generated from the same strict-paired summary data.

Figure 3-E2: Normalized capability contour

Five metrics are normalized to 0-100, where higher is better.

Figure 3-E3: Cost-cache efficiency plane

X-axis is token cache hit rate; Y-axis is observed total cost.

Figure 3-E1: Routing-mode core metric comparison

Infron and OpenRouter are shown side by side by side by routing mode.

Figure 3-E4: E2E latency and Streaming TTFT

Solid lines show E2E latency; dashed lines show Streaming TTFT.

Figure 3-E5: Upstream provider distribution

Provider shares explain cache-domain and performance differences.

Routing-Mode Drill-Down Charts

The horizontal axis shows relative advantage: blue to the right indicates Infron advantage, orange to the left indicates OpenRouter advantage.

Figure 4–E1: Throughput First core metric comparison

Direction and magnitude of relative advantage across five metrics.

Figure 4–E2: Price First core metric comparison

Direction and magnitude of relative advantage across five metrics.

Figure 4–E3: Latency First core metric comparison

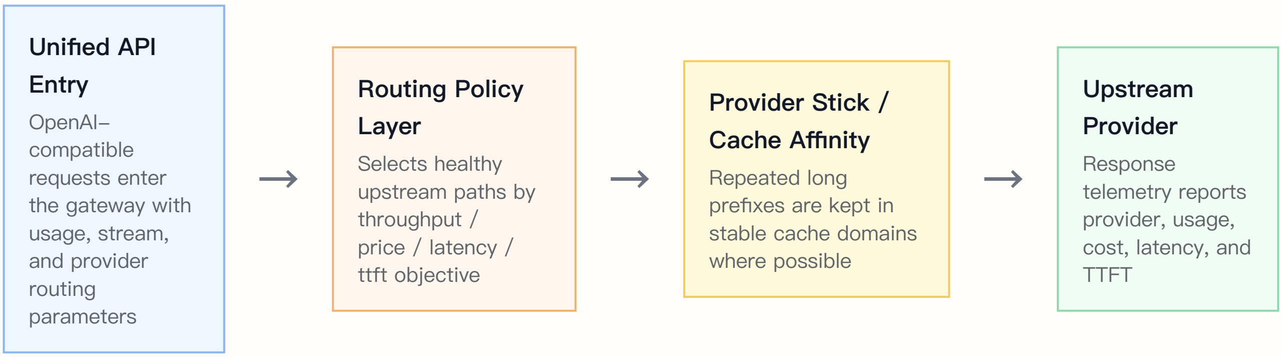
Direction and magnitude of relative advantage across five metrics.

Figure 4–E4: TTFT First core metric comparison

Direction and magnitude of relative advantage across five metrics.

6. Infron Technical Architecture and Cache/Cost Mechanism

Figure 12: Infron Multi-Provider Routing and Cache Control Plane



6.1 Multi-Provider Routing and Observable Control Plane

Requests enter a unified API, and the routing layer selects healthy upstream paths according to throughput, price, latency, or ttft objectives. Whether stable long prefixes stay in the same cache domain directly affects second-request cache-read tokens.

6.2 Provider Stick and Cache Affinity

Provider stick is cache affinity, not a permanent provider lock. Its goal is to reduce cache-domain fragmentation while respecting health and SLA constraints.

6.3 Cost-Control Path

Mechanism	Cache-rate effect	Cost effect	Observable signal
Stable prefix detection	Repeated prefixes are more likely to hit cache	Reduces repeated prefill cost	Payload SHA-256 and second-request cache-read tokens
Provider stick / cache affinity	Reduces cross-domain cache fragmentation	Avoids repeated cache warm-up	Provider distribution and token cache hit rate
Health checks and fallback	Protects availability while sometimes sacrificing cache	Reduces failure cost	HTTP status, provider distribution, tail latency
Cost-aware routing	Prefers lower-cost paths under constraints	Reduces total and per-round cost	Observed cost, cost breakdown coverage, cache-read tokens

7. Provider/Route Drill-Down

Routing mode	Platform	Total requests	Attributed requests	Provider distribution
Throughput First	Infron	384	384	`google-ai-studio` 383, `google-vertex` 1
Throughput First	OpenRouter	384	384	`Google` 382, `Google AI Studio` 2
Price First	Infron	400	400	`google-ai-studio` 400

Routing mode	Platform	Total requests	Attributed requests	Provider distribution
Price First	OpenRouter	400	400	`Google` 322, `Google AI Studio` 78
Latency First	Infron	400	400	`google-ai-studio` 400
Latency First	OpenRouter	400	400	`Google` 304, `Google AI Studio` 96
TTFT First	Infron	400	400	`google-vertex` 400
TTFT First	OpenRouter	400	400	`Google` 275, `Google AI Studio` 125

Upstream Provider Detail Distribution

Routing mode	Platform	Upstream provider	Requests	Share	first/ second	Covered rounds	Avg TTFT	Avg latency	Prompt tokens	Completion tokens	Reas toke
Throughput First	Infron	`google-ai-studio`	383	99.74%	192/191	192	1472.01 ms	1540.31 ms	617396	4596	4280
Throughput First	Infron	`google-vertex`	1	0.26%	0/1	1	1540.52 ms	1634.98 ms	1611	13	13
Throughput First	OpenRouter	`Google`	382	99.48%	191/191	192	1974.58 ms	2194.39 ms	615402	1146	0
Throughput First	OpenRouter	`Google AI Studio`	2	0.52%	1/1	2	2019.89 ms	2045.66 ms	3226	6	0
Price First	Infron	`google-ai-studio`	400	100.00%	200/200	200	1580.24 ms	1649.17 ms	644800	4800	4441
Price First	OpenRouter	`Google`	322	80.50%	163/159	174	2046.50 ms	2210.28 ms	518742	966	0
Price First	OpenRouter	`Google AI Studio`	78	19.50%	37/41	52	2495.23 ms	2516.65 ms	125814	234	0
Latency First	Infron	`google-ai-studio`	400	100.00%	200/200	200	1579.91 ms	1648.06 ms	644800	4800	4447
Latency First	OpenRouter	`Google`	304	76.00%	164/140	171	1983.55 ms	2102.82 ms	489744	912	0
Latency First	OpenRouter	`Google AI Studio`	96	24.00%	36/60	67	2413.04 ms	2438.39 ms	154848	288	0
TTFT First	Infron	`google-vertex`	400	100.00%	200/200	200	1588.25 ms	1694.77 ms	644400	5071	5071

Routing mode	Platform	Upstream provider	Requests	Share	first/second	Covered rounds	Avg TTFT	Avg latency	Prompt tokens	Completion tokens	Reas toke
TTFT First	OpenRouter	`Google`	275	68.75%	139/136	148	2014.54 ms	2182.78 ms	443025	825	0
TTFT First	OpenRouter	`Google AI Studio`	125	31.25%	61/64	73	2516.64 ms	2673.55 ms	201625	375	0

7.1 Cache-Rate and Cost Divergence Drill-Down

This table combines cache, cost, provider distribution, and reasoning telemetry to explain routing-mode differences.

Routing mode	Cache-hit delta	Infron cost multiple	Infron top path	OpenRouter top path	Reasoning token delta	Main attribution
Throughput First	+0.00 pp	1.03x	`google-ai-studio` 99.74%	**`Google` 99.48%**	+4293	Infron has higher cache but higher cost; inspect upstream unit price and completion/reasoning tokens
Price First	+0.00 pp	1.03x	`google-ai-studio` 100.00%	**`Google` 80.50%**	+4441	Infron has higher cache but higher cost; inspect upstream unit price and completion/reasoning tokens
Latency First	+0.00 pp	1.03x	`google-ai-studio` 100.00%	**`Google` 76.00%**	+4447	Infron has higher cache but higher cost; inspect upstream unit price and completion/reasoning tokens
TTFT First	+0.00 pp	**0.78x**	**`google-vertex` 100.00%**	`Google` 68.75%	+5071	Cache is tied; Infron leads cost

8. Stratified Results: Prompt-Length Cache Performance

Prompt-length stratification was not enabled for this run.

9. Stratified Results: Group-Level Stability

Throughput First

Platform	Group	Rounds	Successful rounds	Token cache hit rate	Observed cost	P95 latency	P95 TTFT
Infron	1	42	42	**0.00%**	$0.07051900	**1670.25 ms**	**1623.90 ms**
Infron	2	50	50	**0.00%**	$0.08420000	**1605.70 ms**	**1531.73 ms**

Platform	Group	Rounds	Successful rounds	Token cache hit rate	Observed cost	P95 latency	P95 TTFT
Infron	3	50	50	**0.00%**	$0.08420000	**2053.52 ms**	**1982.92 ms**
Infron	4	50	50	**0.00%**	$0.08420000	**2081.99 ms**	**2026.76 ms**
OpenRouter	1	42	42	**0.00%**	**$0.06841800**	2960.07 ms	2458.08 ms
OpenRouter	2	50	50	**0.00%**	**$0.08145200**	3664.90 ms	2872.34 ms
OpenRouter	3	50	50	**0.00%**	**$0.08145000**	2758.53 ms	2459.16 ms
OpenRouter	4	50	50	**0.00%**	**$0.08145000**	2717.64 ms	2599.87 ms

Price First

Platform	Group	Rounds	Successful rounds	Token cache hit rate	Observed cost	P95 latency	P95 TTFT
Infron	1	50	50	**0.00%**	$0.08420000	**1747.48 ms**	**1677.44 ms**
Infron	2	50	50	**0.00%**	$0.08420000	**1969.58 ms**	**1913.10 ms**
Infron	3	50	50	**0.00%**	$0.08420000	**1793.17 ms**	**1719.47 ms**
Infron	4	50	50	**0.00%**	$0.08420000	**1770.00 ms**	**1694.23 ms**
OpenRouter	1	50	50	**0.00%**	**$0.08147300**	2844.60 ms	2493.02 ms
OpenRouter	2	50	50	**0.00%**	**$0.08150400**	2720.43 ms	2700.36 ms
OpenRouter	3	50	50	**0.00%**	**$0.08145000**	2634.43 ms	2377.10 ms
OpenRouter	4	50	50	**0.00%**	**$0.08145100**	2751.46 ms	2713.27 ms

Latency First

Platform	Group	Rounds	Successful rounds	Token cache hit rate	Observed cost	P95 latency	P95 TTFT
Infron	1	50	50	**0.00%**	$0.08420000	**1786.42 ms**	**1729.53 ms**
Infron	2	50	50	**0.00%**	$0.08420000	**1852.36 ms**	**1779.56 ms**
Infron	3	50	50	**0.00%**	$0.08420000	**1816.45 ms**	**1750.92 ms**
Infron	4	50	50	**0.00%**	$0.08420000	**1767.04 ms**	**1691.21 ms**
OpenRouter	1	50	50	**0.00%**	**$0.08151700**	2852.49 ms	2747.80 ms
OpenRouter	2	50	50	**0.00%**	**$0.08145300**	2619.48 ms	2434.11 ms
OpenRouter	3	50	50	**0.00%**	**$0.08145000**	2791.32 ms	2588.29 ms
OpenRouter	4	50	50	**0.00%**	**$0.08147600**	2542.01 ms	2501.53 ms

TTFT First

Platform	Group	Rounds	Successful rounds	Token cache hit rate	Observed cost	P95 latency	P95 TTFT
Infron	1	50	50	**0.00%**	**$0.06324800**	**2201.85 ms**	**2078.93 ms**
Infron	2	50	50	**0.00%**	**$0.06322000**	**2036.97 ms**	**1828.58 ms**

Platform	Group	Rounds	Successful rounds	Token cache hit rate	Observed cost	P95 latency	P95 TTFT
Infron	3	50	50	**0.00%**	**$0.06323000**	**1973.04 ms**	**1797.50 ms**
Infron	4	50	50	**0.00%**	**$0.06324400**	**1964.06 ms**	**1864.54 ms**
OpenRouter	1	50	50	**0.00%**	$0.08149700	3072.54 ms	2985.50 ms
OpenRouter	2	50	50	**0.00%**	$0.08145000	2813.70 ms	2548.75 ms
OpenRouter	3	50	50	**0.00%**	$0.08145100	2986.73 ms	2438.78 ms
OpenRouter	4	50	50	**0.00%**	$0.08152700	4471.62 ms	3188.64 ms

10. Discussion: Business Value, Boundaries, and Engineering Implications

Business decisions should not rely on one metric. Stable long-context and high-frequency template workloads should prioritize cache rate and cost; realtime interaction must constrain TTFT and E2E latency; batch processing often prioritizes throughput and failure cost.

Routing mode	Business objective	Observed result	Scenarios	Caveat
Throughput First	Maximize output capacity per unit time	Infron leads overall (3/5 metrics)	Batch generation, offline summaries, backend processing	Decide with budget, SLA, and cache stability together
Price First	Minimize request and token cost	Infron leads overall (3/5 metrics)	High-frequency templates, support automation, marketing, RAG prefixes	Decide with budget, SLA, and cache stability together
Latency First	Minimize full-response waiting time	Infron leads overall (3/5 metrics)	Online chat, agent chains, IDE/writing assistants, realtime tools	Decide with budget, SLA, and cache stability together
TTFT First	Minimize streaming first-token time	Infron leads overall (4/5 metrics)	Streaming chat, realtime copilots, first-screen feedback	Decide with budget, SLA, and cache stability together

11. Conclusion

Routing-Mode Conclusions

Routing mode	Objective	Cache winner	Cost winner	Throughput winner	E2E latency winner	TTFT winner	Interpretation
Throughput First	Maximize output capacity per unit time	**Tie**	**OpenRouter**	**Infron**	**Infron**	**Infron**	Infron leads overall (3/5 metrics)
Price First	Minimize request and token cost	**Tie**	**OpenRouter**	**Infron**	**Infron**	**Infron**	Infron leads overall (3/5 metrics)

Routing mode	Objective	Cache winner	Cost winner	Throughput winner	E2E latency winner	TTFT winner	Interpretation
Latency First	Minimize full-response waiting time	**Tie**	**OpenRouter**	**Infron**	**Infron**	**Infron**	Infron leads overall (3/5 metrics)
TTFT First	Minimize streaming first-token time	**Tie**	**Infron**	**Infron**	**Infron**	**Infron**	Infron leads overall (4/5 metrics)

12. Limitations, Missing Data, and Next Experiments

Limitation	Impact	Next step	Current handling
Full routing trace	Cannot prove every provider choice, fallback, and retry path hop by hop	Add provider routing trace, decision logs, and fallback reasons	Use only returned provider fields and provider distribution
Longer time window	4x50 observes short-window stability but not day-level drift	Add soak tests and repeated windows	Scope conclusions to this run
Production corpus	Built-in templates do not cover every workload distribution	Use sanitized production-stratified corpora	Discuss representative long-context templates only
Cost-field consistency	Cost coverage and semantics may differ by platform	Reconcile with billing and provider cost breakdown	Use only explicitly returned cost fields

13. Reproducibility Appendix

Artifact	Path
Summary	`export/ gemini3_flash_preview_routing_sort_cache_cost_ab_4x50_stream_ttft_reasoning_default_length_ summary.json`
Paired dataset	`export/ gemini3_flash_preview_routing_sort_cache_cost_ab_4x50_stream_ttft_reasoning_default_length_ benchmark_pairs.csv`
Request-level dataset	`export/ gemini3_flash_preview_routing_sort_cache_cost_ab_4x50_stream_ttft_reasoning_default_length_ benchmark_requests.jsonl`
Filtered structured records	`export/ gemini3_flash_preview_routing_sort_cache_cost_ab_4x50_stream_ttft_reasoning_default_length_ records.json`
Excluded-record audit	`export/ gemini3_flash_preview_routing_sort_cache_cost_ab_4x50_stream_ttft_reasoning_default_length_ records_excluded.json`
Test source	`tests/test_rerun_routing_sort_cache_cost_ab.py`

Artifact	Path
Benchmark runner source	`scripts/rerun_routing_sort_cache_cost_ab.py`
HTML report renderer source	`scripts/render_glm52_deepseek_style_report.py`
Dataset reference	`business_representative` built-in representative business templates; request-level export is `benchmark_requests`